# Day 4: Academic Pathways

## How can your BYU education prepare you for the work you want to do?

DS 115: Exploring Data Science

# Where we left off

## What did the job postings tell us about the skills people use?

Recall one responsibility and one skill. Compare with a neighbor.

Different jobs draw on different combinations of **computing, statistics, mathematics, communication, and context**.

# Today’s question

## How can these academic pathways help you grow into data science?

You can begin learning while you are still figuring out your direction.

# What you should leave with

- An understanding of **why the DS degree includes different disciplines**.
- A way to compare majors using **their actual requirements**.
- Resources to help you **begin your graduation plan and get advice**.

# The curriculum helps you learn the work

You are here to develop these abilities. You do not need to arrive with them.

| As you move through the degree… | You learn how to… |
| --- | --- |
| Mathematics | Understand the structure behind models |
| Statistics | Evaluate evidence and uncertainty |
| Computing | Work with data and build reliable programs |
| Courses that bring them together | Use these foundations in real investigations |

# The structure of the Data Science BS

| Part of the degree | What it develops |
| --- | --- |
| Introductory courses | A starting vocabulary and basic tools |
| Mathematical foundations | A language for models and computation |
| Statistical foundations | Reasoning about evidence and uncertainty |
| Computing foundations | Reliable programs and organized data |
| Data science core | Practice combining the foundations |
| Electives and specialization | More depth in areas you want to explore |
| Advanced theory and experience | Deeper models, capstone work, or research |

View the actual Data Science BS requirements.

# Requirement 1: Introductory courses

| Courses listed in the introductory block | Starting point |
| --- | --- |
| DS 115 | Explore the field and possible pathways |
| DS 180 | Begin working with data science ideas |
| CS 111 | Build a programming foundation |
| MATH 112 | Develop calculus tools |
| STAT 121 | Begin statistical reasoning |

Use your degree audit and advisor to confirm how prior credit and introductory requirements apply to you.

# Requirement 2: Mathematical foundations

**MATH 113, MATH 213, MATH 215**

## How does a model combine information from many variables?

Calculus helps us describe change. Linear algebra helps us represent and compute with many quantities at once.

Think back to using several student characteristics to predict GPA.

# Requirement 3: Statistical foundations

**STAT 230, STAT 240, STAT 251, STAT 330**

## What does the evidence support, and how uncertain are we?

- Study design and statistical modeling.
- Probability, inference, and Bayesian reasoning.

A relationship in the GPA data did not automatically tell us what causes student success.

# Requirement 4: Computing foundations

**CS 235, CS 240, CS 452**

## How will you make the work reliable when the data and code get larger?

- Data structures and algorithms.
- Software design and testing.
- Database organization and access.

Think back to the pipelines and systems in the Day 3 postings.

# Requirement 5: Data science core

| Course | A question it helps you approach |
|---|---|
| CS 270 | How can a model learn from examples? |
| CS 404 | How should technology affect people? |
| MATH 380 | What mathematics supports data science? |
| PHSCS 383 | How does knowledge of a system guide reasoning? |
| STAT 281 | How can a visual communicate evidence? |
| STAT 386 | How do we carry out an analysis with data? |

# Requirement 6: Electives and specialization

The degree includes **12 hours of approved electives**.

| A direction you could explore | Examples from the approved list |
|---|---|
| Deeper modeling and theory | Advanced ML, deep learning, probability and inference |
| Mathematical methods | Numerical linear algebra and model uncertainty |
| Data science in other fields | Bioinformatics, economics, and geoscience |
| Work on larger projects | Capstone coursework and approved internship credit |

You can use these choices to develop interests you discover along the way.

# Requirement 7: Advanced theory and experience

Choose **3 hours from the approved list**.

| Some options deepen theory and methods | Others emphasize experience |
|---|---|
| CS 473: Advanced Machine Learning | CS 483: Data Science Capstone 2 |
| CS 474: Deep Learning | Approved undergraduate research |
| MATH 522: Mathematics of Deep Learning | Approved internship credit |
| STAT 486: Machine Learning | Approved physics capstone or research |

A course used here cannot also count toward Requirement 6.

# Different majors, different priorities

We have seen how the Data Science BS combines these foundations. How do other programs arrange them?

The requirements show what each pathway asks you to practice in greater depth.

# Questions for reading a degree’s requirements

- Which foundations does it share with Data Science?
- Where does it build more depth?
- Where do students have a choice?

We will look at the requirements together. You can notice differences without deciding on a major today.

# Five data science pathways at BYU

- Data Science BS
- Statistics BS, Data Science emphasis
- Machine Learning BS
- Mathematics BS, Applied and Computational Mathematics emphasis (ACME)
- Applied Physics BS, Data Science emphasis

We will look through the actual program requirements together.

# A shared foundation

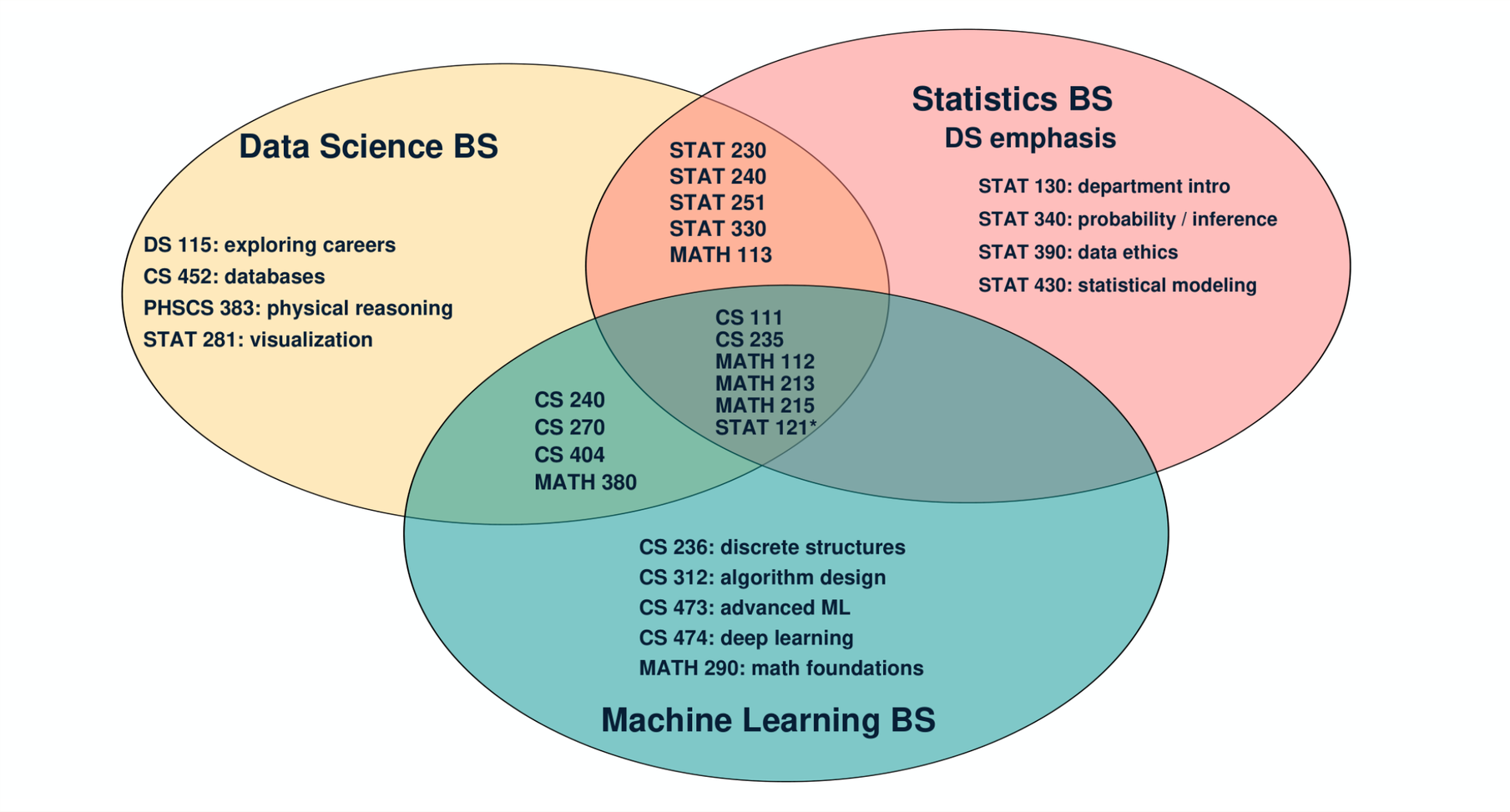
**Data Science BS**, **Statistics BS, Data Science emphasis**, and **Machine Learning BS** share substantial early coursework.

## CS 111, CS 235, MATH 112, MATH 213, MATH 215, STAT 121*

*Machine Learning allows STAT 201 as an alternative to STAT 121.

Learning these foundations gives you time to discover which parts of the field you enjoy.

# Overlapping foundations, different depth




Selected core requirements. *ML allows STAT 201 instead. A course outside a program's core may still be an elective.

# A starting point when you are unsure

## If you are unsure which pathway to pursue, start with the Data Science BS.

Its broad foundation lets you experience mathematics, statistics, and computing as you learn about the field.

As your interests become clearer, you can stay in DS or explore a related major with an advisor.

# Your major matters, and so does how you use it

## Your success in data science is unlikely to hinge on which of these related majors you choose.

Your major shapes the foundation you build. **How you learn and what you do with that foundation matter too.**

Courses, projects, research, and work experience can help you grow in directions beyond your required curriculum.

# A MAP sheet gives you a starting sequence

## Data Science BS: Major Academic Plan (MAP) sheet

- Lists the degree requirements.
- Suggests **which courses to take and when**.
- Places major courses alongside GE and religion requirements.

Use this as a starting point for your graduation-plan assignment.

Adapt it to your prior credit, responsibilities, and applicable catalog year.

# Prerequisites affect the order

| A standard computing route | What needs to come first |
| --- | --- |
| CS 111 | Check placement and preparation |
| CS 235 | CS 111 is one accepted prerequisite option |
| CS 240 | CS 235, or instructor consent |

A course can be required and still be too early to take next semester.

# Prerequisites and offerings affect the sequence

**Hypothetical example:** no prior credit or summer courses. Each prerequisite must finish before the next course begins.

| Course | Prerequisite | Offered |
|---|---|---|
| A: Foundations | None | Fall and Winter |
| B: Intermediate methods | A | Fall and Winter |
| C: Advanced project | B | Winter only |

## If you start A in Fall of Year 1, when can you first finish C?

A in Fall 1, B in Winter 1, C in Winter 2. Starting A in Winter 1 still reaches C in Winter 2.

# MyMAP helps you see your own progress

## Open MyMAP

- See how your completed courses apply to requirements.
- Identify requirements you still need.
- Organize a tentative course plan.

The MAP sheet gives a suggested sequence. MyMAP helps you connect it to your own record.

# An advisor can help you move forward

**CMS Advisement Center**

An advisor can help you understand prior credit, plan a course sequence, and explore a possible change of major.

## You can ask for help before you have everything figured out.

Visit the advisement page for appointment options. You can also contact the center at **801-422-2674** or cmsadvising@byu.edu.

# Resources for the graduation-plan assignment

1. Use the **catalog** to identify your requirements.
2. Use the **MAP sheet** to help arrange the course sequence.
3. Use **MyMAP** to account for your own credit and remaining work.
4. Ask an **advisor** about questions or changes you want to explore.

Your assignment is a plan from now through graduation. Use Canvas for the format, deadline, and submission instructions.

# Exit ticket

1. What did you notice about the overlap among the majors?
2. Name one course and something it will help you learn.
3. Which resource will help you take your next step?

You can still be undecided about your eventual major.

# You can start before you decide

## You do not need to find the perfect major before you begin.

If you are unsure, begin in **Data Science** with basic **mathematics, programming, and statistics**. Notice which kinds of work resonate with you.

## Let what you learn help you refine your choice.

These pathways share many early foundations, so there is often room to pivot as your interests become clearer.

# Reference: planning resources

- Data Science BS catalog
- Data Science MAP sheet
- MyMAP
- CMS Advisement Center
- Class schedule

# Reference: credit totals

| Program | Published major credit hours |
| --- | --- |
| Data Science BS | 74.5 |
| Statistics BS, Data Science emphasis | 58 |
| Machine Learning BS | 74 |
| Mathematics BS, ACME emphasis | 70-76 |
| Applied Physics BS, Data Science emphasis | 66-68 |

Major credits alone do not determine time to graduation. Account for prior credit, university requirements, prerequisites, and offerings.

# Reference: questions for an advisor

- How does my prior credit apply to this major?
- Which course sequence should I begin with?
- What would change if I switched to a related major?
- What could I add through electives or experience?

Bring your questions, even if your plan is still a rough draft.